5-TIER CYBER-PHYSICAL SYSTEM ARCHITECTURE DIAGRAM

AI-Powered Smart Campus Resource Management System | Course: E1CSA313 | Batch: DP1-24-28-CSE-GEN-L-01

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TIER 5: HUMAN-MACHINE INTERACTION & ACTUATION DOCK (CLIENT)

Zero-install responsive frontend delivering multi-role views, live spatial telemetry & sub-second override controls

- Central Multi-Role Dashboard
- Interactive Building Drawer
- Dynamic D3 Sankey Power Flow
- **Manual Actuator Override Dock**

TIER 4: 3D WEBGL GRAPHICS & SPATIAL SHADER ENGINE

Procedural Blender geometry pipeline rendering campus assets at 60 FPS under 15 MB VRAM footprint

- Three.js r170 Scene Graph
- OrbitControls & Frustum Culling
- Dynamic Thermal Shaders
- Low-Poly Parametric GLB <15MB

TIER 3: COMPUTATIONAL INTELLIGENCE & DECISION ENGINE

Core algorithmic layer executing space reclamation, microgrid peak-shaving dispatch & timetable constraint solving

- **Timetable CSP Solver**
  Eliminates 100% room clashes
- **15-Min Hysteresis Buffer**
  Zero false-alarm chiller alerts
- **1.2 MWh BESS Dispatch**
  ToD peak tariff shaving (>18%)
- **Thermal Euler Differential Solver**
  Lumped-capacitance building dynamics

TIER 2: IOT TELEMETRY & REAL-TIME DATA INGESTION ENGINE

Physics-coupled state engine synthesizing 1,000+ virtual sensor beacons, ERP schedules & weather inputs

- 1,000+ Spatial Sensor Beacons
- Alliance ERP Timetable Scraper
- BESCOM Tariff & Meter Feeds
- WebSocket State Hydration <100ms

TIER 1: PHYSICAL CAMPUS INFRASTRUCTURE & ELECTRICAL ASSETS

Alliance University 60-acre Anekal campus buildings, renewable solar fields & power distribution equipment

- Admin & Library Blocks
- ACED Labs & Classrooms
- 480 kWp Rooftop Solar PV
- 1.2 MWh LiFePO4 BESS
- Central Chiller Plant

DESIGNED BY: LIKITH S (LEAD) | MENTOR: DR. RASHMI MOTHKUR

DATA FLOW DIAGRAMS: LEVEL-0 (CONTEXT) & LEVEL-1 (PROCESS FLOW)

Cyber-Physical Data Pathways · Space Reclamation & Microgrid Load Optimization

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DFD LEVEL-0: SYSTEM CONTEXT DIAGRAM



DFD LEVEL-1: SUB-PROCESS PIPELINE



DRAWING NO: AU-DP1-DWG-002 | SHEET 2 OF 5 | SCALE: N.T.S. | PROJECTION: THIRD ANGLE

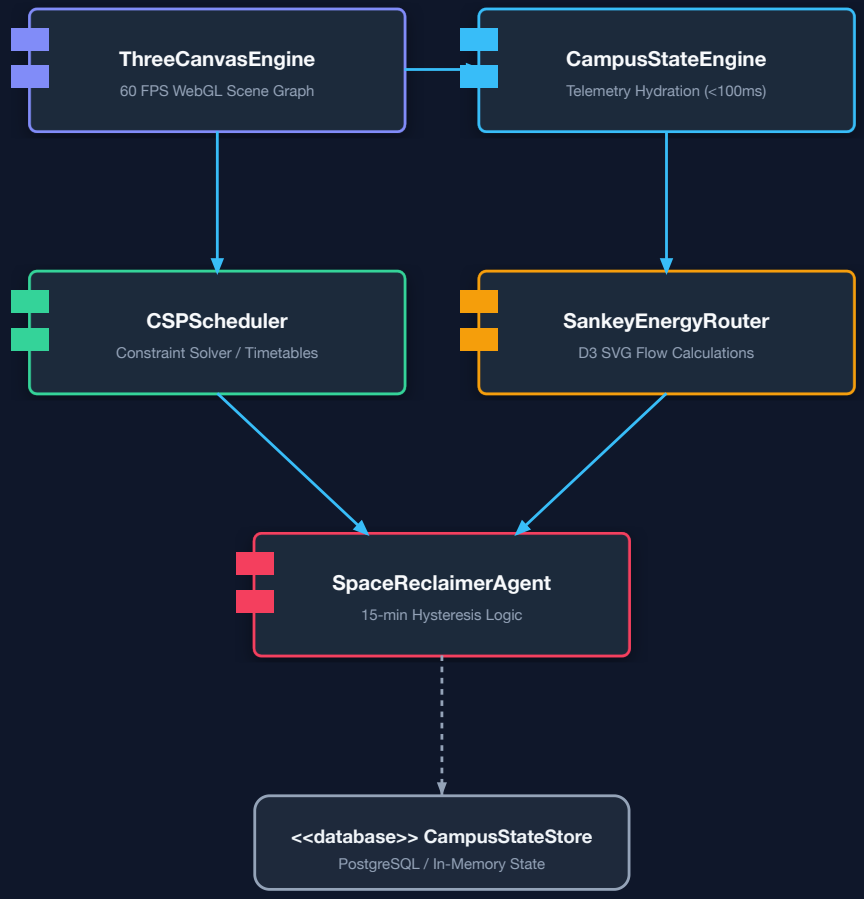
DESIGNED BY: LIKITH S (LEAD) | MENTOR: DR. RASHMI MOTHKUR

UML SPECIFICATION: COMPONENT DECOMPOSITION & SEQUENCE LIFECYCLE

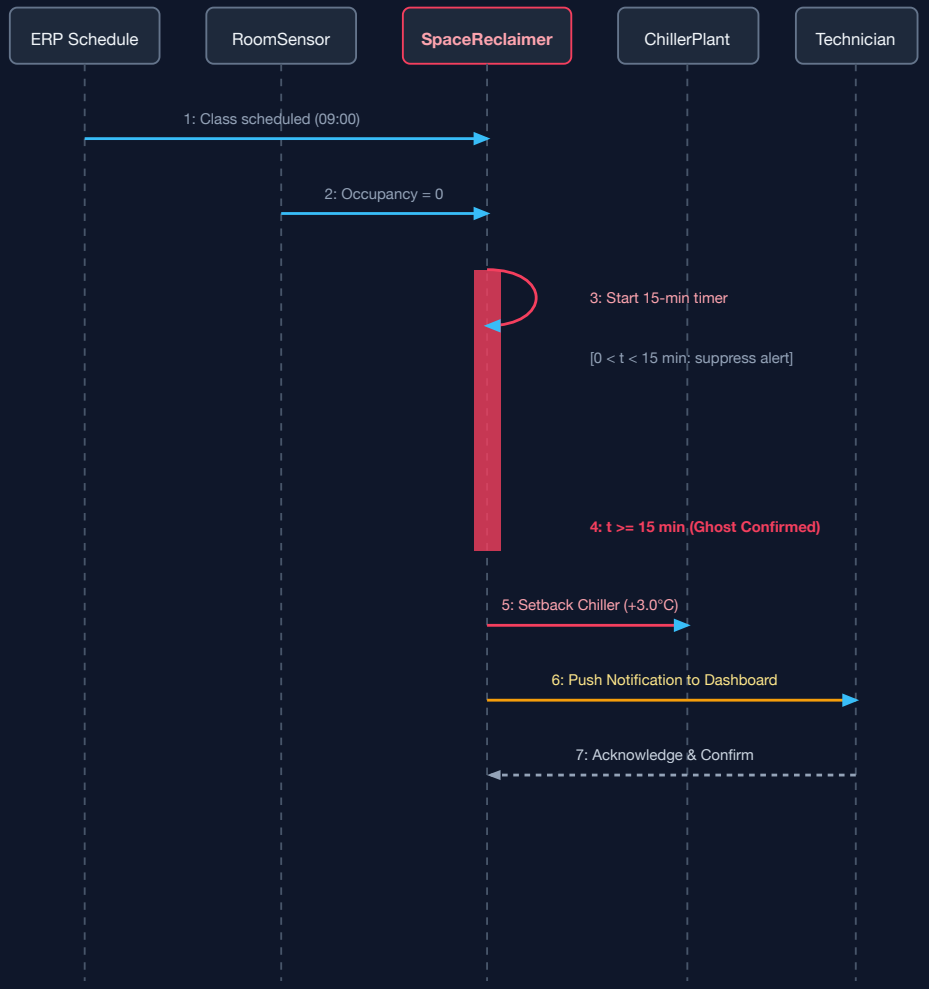
Software Engineering Contracts & 15-Minute Hysteresis Sequence Lifecycle

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<<component>> SUBSYSTEM DECOMPOSITION



SEQUENCE: 15-MIN HYSTERESIS GHOST-BOOKING LIFECYCLE

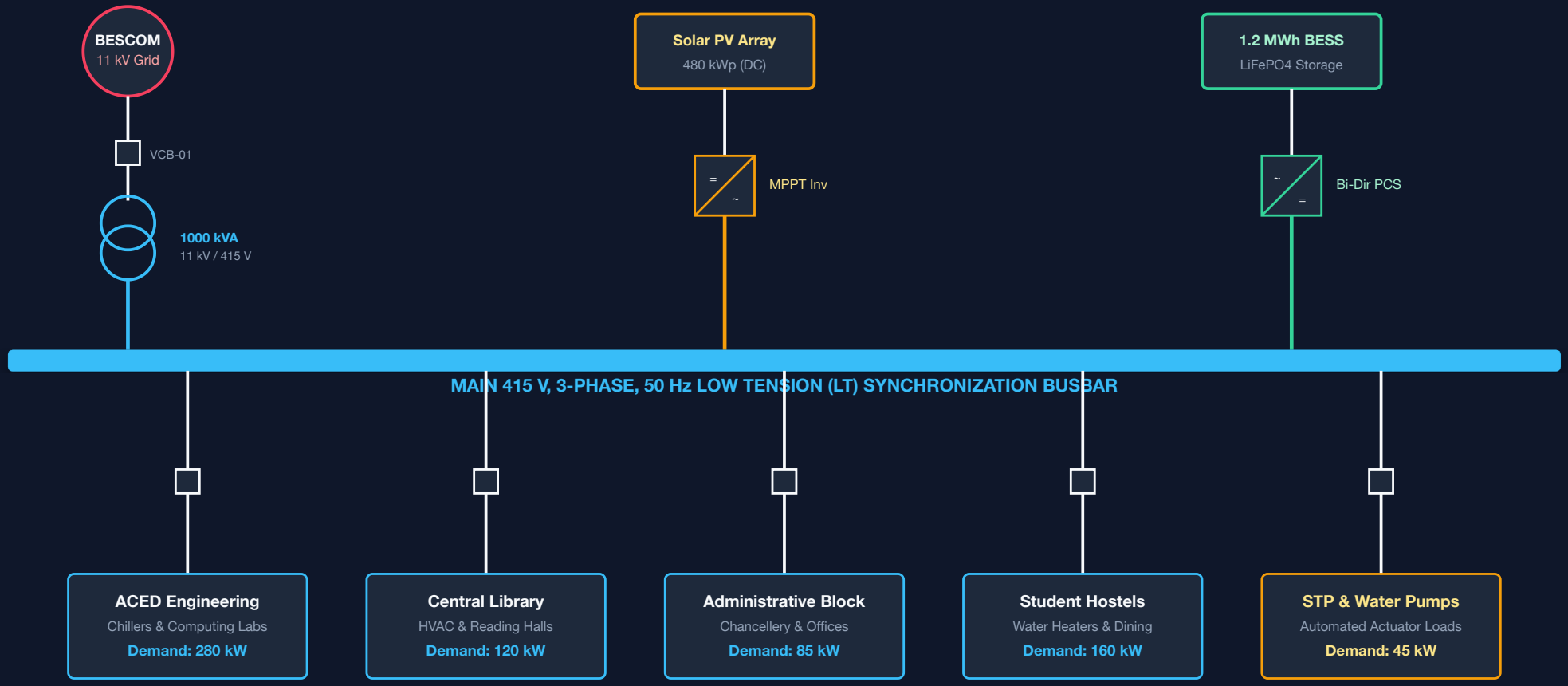


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CAMPUS MICROGRID SINGLE-LINE DIAGRAM (SLD) & POWER FLOW

11 kV BESCOM Grid Incomer · 480 kWp Solar PV · 1.2 MWh BESS Storage · 415 V LT Distribution

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DRAWING NO: AU-DP1-DWG-004 | SHEET 4 OF 5 | SCALE: N.T.S. | PROJECTION: THIRD ANGLE

DESIGNED BY: LIKITH S (LEAD) | MENTOR: DR. RASHMI MOTHKUR

ALLIANCE UNIVERSITY ANEKAL CAMPUS: 2D MASTER SITE PLAN & ASSET LAYOUT

60-Acre Coordinate Geometry · Building Footprints · Rooftop Solar PV · Microgrid Substation Yard

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ADMINISTRATIVE BLOCK

Chancellery & Admissions (3 Floors)

[Coordinates: 0, -18m | 9m x 7m Footprint]

CENTRAL LIBRARY

Knowledge Hub & Rotunda (4 Floors)

[Coordinates: 0, 0m | 8.6m x 8.6m]

ACED ENGINEERING BLOCK

Computing Labs & Lecture Halls (7 Floors)

[Coordinates: +24m, +8m | 18 Classrooms]

Rooftop Solar PV: 240 kWp

11 kV SUBSTATION & BESS YARD

1000 kVA VCB

1.2 MWh BESS

CHILLER PLANT & STP

Central HVAC Pumping Station

[Automated Setback Actuators]

STUDENT HOSTELS & DINING

Residential Blocks (6 Floors)

Rooftop Solar PV: 240 kWp

DRAWING NO: AU-DP1-DWG-005 | SHEET 5 OF 5 | SCALE: 1:1000 | PROJECTION: THIRD ANGLE

DESIGNED BY: LIKITH S (LEAD) | MENTOR: DR. RASHMI MOTHKUR